

CryoSand: An Open Trade-Space Sandbox for In-Space Cryogenic Propellant Storage

First Written Report (Progress Report)

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Abstract

CryoSand is an open tool for the in-space cryogenic storage trade: when does it pay to refrigerate a propellant tank instead of only insulating it? This report covers the first month of the project. The physics stack (layers L0–L5) has been specified, and build steps 1–5 are implemented with 18 analytic-limit tests passing. A first trade has been run. Four findings stand out. (i) The optimal cryocooler lift is always either zero or the full heat leak, never in between. (ii) For a vented tank with constant specific masses, the crossover duration has the closed form $t^* = h_{fg} \mu_{\text{act}}$ and does not depend on insulation, tank size or environment. (iii) With the Strobridge cooler-mass correlation, a baseline LEO tank gives $t^* \approx 57$ days for LH_2 and ≈ 1.6 days for LO_2 . This is the same order as the published LEO estimates of ~ 64 and ~ 5 days. (iv) Closing the tank shifts the crossover by exactly the hold time. An indicative bound calculation puts the LH_2 crossover anywhere from 57.2 to 409 days, depending on the ullage closure. So the well-mixed assumption can move the crossover, not just the vent schedule.

1 Problem and objectives

A propellant tank in orbit holds liquid at 20–112 K behind an outer surface at roughly 180–260 K, so heat always leaks in. Each watt that reaches the liquid evaporates it at $\dot{m} = \dot{Q}/h_{fg}$ (for LH_2 , 1 W boils off ≈ 0.19 kg/day). A designer can spend mass on insulation and accept the boil-off, or on a cryocooler plus its power system and radiator to remove the heat. The mission duration at which the second option becomes lighter is the *passive/active crossover* t^* . No open tool maps it; the fast storage codes that could be licensed [3].

The project has two objectives: **(E) engineering:** publish a map of t^* across propellant, tank scale, environment and duration; **(R) research:** determine whether the near-universal *well-mixed ullage* assumption moves t^* or only changes the vent schedule. Objective R is answered by a *bounding argument*. Two analytic closures bracket the real pressure rise, and both are carried through the full trade. The spring CFD campaign then refines where real tanks sit between the bounds, but the result does not depend on it.

2 Physical model

The model follows each watt from the Sun to the liquid in six independently testable layers (SI throughout; fluid properties only from CoolProp [2]).

L0 Environment. The outer surface absorbs sunlight, Earth albedo and Earth IR, and radiates to a 4 K sink. Averaged over a sphere’s area,

$$q_{\text{abs}} = \alpha \left(\frac{1}{4} G_{\odot} + F_E a G_{\odot} \right) + \varepsilon F_E G_{\text{IR}}, \quad \varepsilon \sigma (T_s^4 - T_{\text{sink}}^4) = q_{\text{abs}}, \quad (1)$$

where $F_E = \frac{1}{2} \left[1 - \sqrt{1 - (R_E/(R_E + h))^2} \right]$ is the sphere-to-Earth view factor. The inward heat leak is less than 1% of q_{abs} , so the outer-surface temperature T_s is set by radiative equilibrium.

L1 Insulation. The multilayer-insulation (MLI) blankets are resistances in series. Struts and plumbing bypass the blankets, so they form a *parallel* conductance:

$$\dot{Q}_{\text{leak}} = \frac{T_s - T_{\text{sat}}}{\sum_i t_i / (k_{\text{eff},i} A) + R_{\text{wall}}} + G_{\text{strut}} (T_s - T_{\text{sat}}), \quad G_{\text{strut}} = \frac{A_s}{L_s} \bar{k}, \quad \bar{k} = \frac{1}{\Delta T} \int k dT. \quad (2)$$

The effective conductivity k_{eff} lumps together radiation, spacer conduction and residual-gas conduction. It is held constant at this fidelity.

L2 Wall. Fourier conduction $\nabla \cdot (k \nabla T) = 0$ is solved by the Summer-2026 finite-difference solver, which uses harmonic-mean face conductivities $k_f = 2k_P k_N / (k_P + k_N)$. At this stage it enters the network as its 1-D limit, $R_{\text{wall}} = t / (kA)$.

L3 Ullage and liquid. Three closures, all first-class:

$$\text{vented: } \dot{m} = \dot{Q}_{\text{net}} / h_{fg}, \quad (3)$$

$$\text{homogeneous (slow bound): } \dot{u} = \dot{Q}_{\text{net}} / m, \quad \rho = m / V, \quad P = P(\rho, u), \quad (4)$$

$$\text{surface (fast bound): } \dot{T}_i = \dot{Q}_{\text{net}} / (m_{\text{surf}} c_p), \quad P = P_{\text{sat}}(T_i). \quad (5)$$

The homogeneous closure is the fully mixed tank, which gives the slowest possible pressure rise. The surface closure puts all incoming energy into an interface layer of mass m_{surf} . Because saturation pressure follows the interface temperature, this gives the fast rise seen in stratified tanks. m_{surf} is swept. With $\dot{Q}_{\text{net}}$ constant, each closure is summarised by the energy E_{store} it can absorb before reaching the vent pressure, which gives the hold time $t_{\text{hold}} = E_{\text{store}} / \dot{Q}_{\text{net}}$.

L4 Cooling. The Carnot coefficient of performance is $\text{COP}_C = T_c / (T_h - T_c)$. The specific power is $\phi = W_{\text{in}} / \dot{Q}_{\text{lft}} = 1 / (\eta \text{COP}_C)$, where η is the fraction of Carnot achieved. The radiator rejects $\dot{Q}_{\text{lft}} + W_{\text{in}}$, not $\dot{Q}_{\text{lft}}$ alone: $A_{\text{rad}} = \dot{Q}_{\text{lft}} (1 + \phi) / [\varepsilon \sigma (T_r^4 - T_{\text{sink}}^4)]$. Cooler mass follows the Strobbridge correlation as scaled in [1], $m_{\text{cool}} = 0.2 \dot{Q}^{0.7} [(T_h - T_c) / T_c]^{1.45}$.

L5 Closure. $m_{\text{pen}} = m_{\text{ins}} + m_{\text{cool}} + m_{\text{pow}} + m_{\text{rad}} + m_{\text{lost}}$. The crossover is the duration at which the minimum of m_{pen} over the design space switches from passive to active.

3 Implementation status

The code is in Python and follows three conventions: pure physics functions, frozen dataclasses for state, and a single CoolProp property wrapper. Physics functions receive property values as arguments, so every analytic limit can be tested without a property library. Table 1 lists the build order.

4 Preliminary results

Baseline: a cylinder with hemispherical ends, $r = 2$ m, cylindrical length 3 m ($V = 71.2$ m³, $A = 88.0$ m²), 90% fill at 1 atm. The insulation design unit is a 10-layer MLI sub-blanket. Saturated properties are the reference values of Table 3 (CoolProp run pending). All specific masses are the provisional register values of Table 3; the numbers below show the structure of the trade and are not final predictions.

4.1 Environment, heat leak and wall

The absorbed flux (with $\alpha/\varepsilon = 0.32/0.86$) and the resulting surface temperature are 219 W/m² and 259 K in LEO at 400 km ($F_E = 0.331$), 109 W/m² and 217 K in deep space at 1 AU, and 47 W/m² and 176 K in Mars transfer. The wall resistance is $R_{\text{wall}} = 2.3 \times 10^{-6}$ K/W, or 7.2×10^{-7}

Table 1: Build order and status. “Written” means the code exists but its tests need CoolProp and have not yet run.

Step	Module	Status / verification
1–2	core, network	Done. Series chain matches the hand sum to 10^{-10} ; $k_f(2, 6) = 3$ exactly.
3	L0, L1	Done. View-factor limits $\{\frac{1}{2}, 0\}$; blankets scale as $1/n$; struts verified to be parallel.
4–5	L4, L5, L3 vented	Done. Carnot limits (14, 2.33 W/W); radiator rejects lift plus work; ZBO balance; zero leak gives exactly zero boil-off.
6	model (steady)	Done. First end-to-end run; brute-force optimum reproduces analytic t^* (§4.3).
7	L3 closed bounds	Written. Ordering, collapse and energy-closure tests written (need CoolProp).
8	L2 port	1-D form and cylinder→plane limit done; 2-D port pending.
9–10	trade , app	First crossover sweep done (Fig. 3); Streamlit interface not started.

of a single 10-layer sub-blanket (3.16 K/W), even with a pessimistic $k = 10$ W/m·K. It is *computed* to be negligible, not assumed so. At the LH₂ ZBO optimum (60 layers, $\dot{Q}_{\text{leak}} = 13.8$ W), the struts carry $\approx 9\%$ of the leak with the current placeholder conductance.

4.2 The cooler asymmetry

Figure 1 shows the thermodynamic core of the trade. At the assumed fractions of Carnot, lifting 1 W costs 184 W of electrical power at 20 K but only 11.6 W at 90 K and 8.4 W at 112 K. Every 100 W lifted from LH₂ needs 44.7 m² of 300 K radiator, compared with 3.1 m² for LO₂.

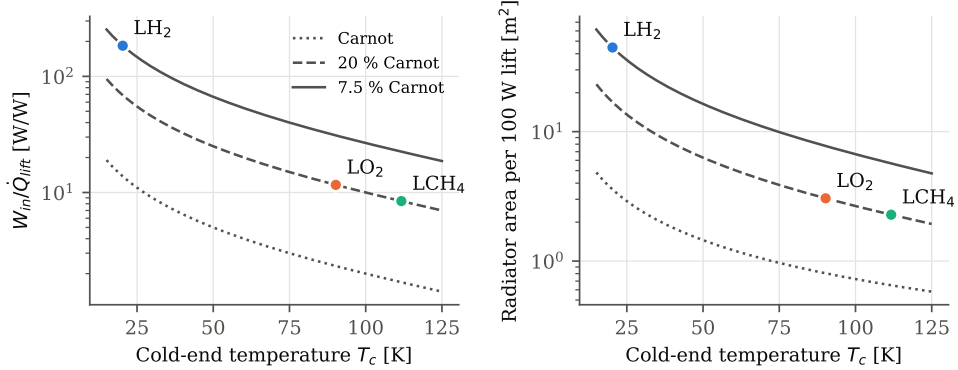


Figure 1: Specific power (left) and radiator area per 100 W of lift (right) against cold-end temperature, for $T_h = T_r = 300$ K and $\varepsilon_r = 0.9$. Markers show each propellant at its assumed fraction of Carnot.

4.3 Structure of the trade and the vented crossover

End-point optimality. For a fixed insulation choice, $m_{\text{pen}}(\dot{Q}_{\text{lift}})$ is the sum of terms that are linear in $\dot{Q}_{\text{lift}}$ (power, radiator, and the boil-off of the uncovered leak) and the concave cooler mass $\propto \dot{Q}_{\text{lift}}^{0.7}$. A concave function on an interval has its minimum at an end point. So the best design is always either fully passive ($\dot{Q}_{\text{lift}} = 0$) or full ZBO ($\dot{Q}_{\text{lift}} = \dot{Q}_{\text{leak}}$). Partial refrigeration is never mass-optimal in this model.

Closed form for constant specific masses. Leaving one watt of leak to boil for a time t costs $\mu_{\text{pas}} = t/h_{fg}$ kg. Removing that watt with a cooler costs $\mu_{\text{act}} = s_{\text{cool}} + s_{\text{pow}}\phi + s_{\text{rad}}(1 + \phi)/q_{\text{rad}}$ kg. Both costs multiply the same leak $\dot{Q}_{\text{leak}}(n)$, so passive wins at *every* insulation thickness exactly

when $\mu_{\text{pas}} < \mu_{\text{act}}$:

$$t^* = h_{fg} \mu_{\text{act}}. \quad (6)$$

With constant specific masses, t^* is therefore independent of insulation, tank size and environment; it depends only on the propellant and the refrigeration chain. A brute-force optimisation over n that includes the parallel strut path reproduces Eq. (6) within 2% in the test suite.

With the Strobbridge correlation. Cooler specific mass now falls as $\dot{Q}^{-0.3}$, so larger leaks buy cheaper watts. Figure 2 shows the baseline LEO case: $t^* = 57$ days for LH_2 (ZBO system 256 kg) and 1.6 days for LO_2 . LCH_4 gives 2.7 days. Plachta and Kittel [1] found ZBO saving mass beyond ~ 64 days for LH_2 and from as little as ~ 5 days for LO_2 in LEO. The LH_2 agreement is close. The LO_2 result is about 3 times shorter; their MLI model (Lockheed correlation $\times 1.8$) and heat-rejection assumptions differ from ours, and reconciling the two is a register task. Figure 3 is the first crossover map. t^* falls with scale ($70 \rightarrow 48$ days for LH_2 , $2.2 \rightarrow 1.4$ days for LO_2 , $r = 0.75 \rightarrow 5$ m). The environment changes it by at most 12%, which is what Eq. (6) predicts: scale enters only through the weak $\dot{Q}^{-0.3}$ economy of scale.

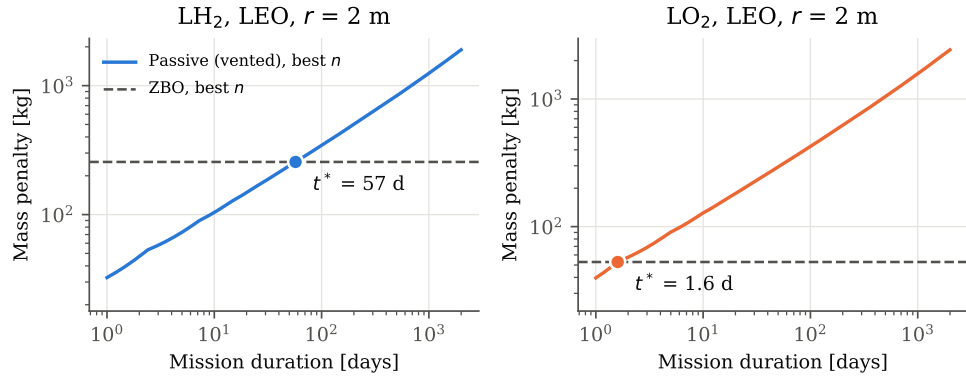


Figure 2: Minimum mass penalty against mission duration for the vented passive design and the ZBO design, each optimised over insulation. The marker is the crossover t^* .

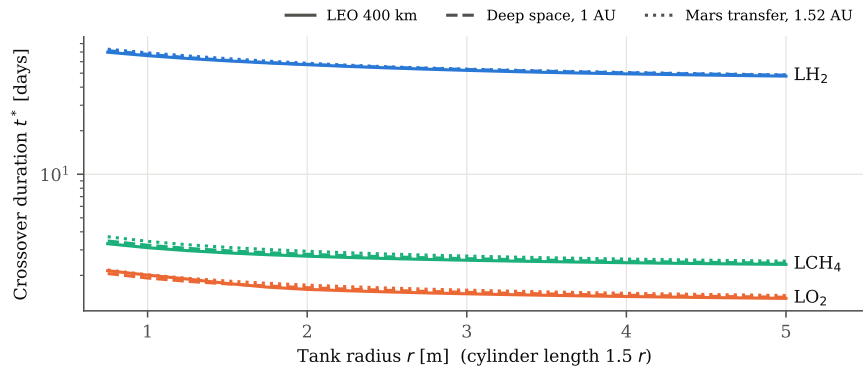


Figure 3: First crossover map (vented closure): t^* against tank radius for three propellants and three environments. The line style marks the environment.

4.4 Closed tank: the ullage closure moves the crossover

A closed tank absorbs E_{store} before its first vent, so for a given insulation choice $m_{\text{lost}} = (\dot{Q}_{\text{leak}} t - E_{\text{store}})^+ / h_{fg}$. Repeating the argument above gives

$$t_{\text{closed}}^* = t_{\text{vented}}^* + E_{\text{store}} / \dot{Q}_{\text{leak}} = t_{\text{vented}}^* + t_{\text{hold}}. \quad (7)$$

Eq. (7) turns objective R into a direct comparison: *the ullage assumption matters exactly when the spread in hold time between the two bounds is comparable to t^** . Figure 4 and Table 2 evaluate this for venting at 3 bar. The homogeneous bound gives hold times of 221 days (LH₂) and 324 days (LO₂). Both are longer than the vented t^* . A conduction-only interface layer (§5, item 1) gives 18 and 21 days. The resulting crossover spans 57.2–409 days for LH₂ and 1.6–943 days for LO₂. These numbers are indicative: they use approximate liquid c_p and k [7] and a Clausius–Clapeyron $T_{\text{sat}}(3 \text{ bar})$ until CoolProp replaces them. Even so, the spread is a factor of about 7 for LH₂ and several hundred for LO₂, so the well-mixed assumption can plausibly move the crossover, not only the vent schedule. Confirming or refuting this with the exact closures is now the first priority.

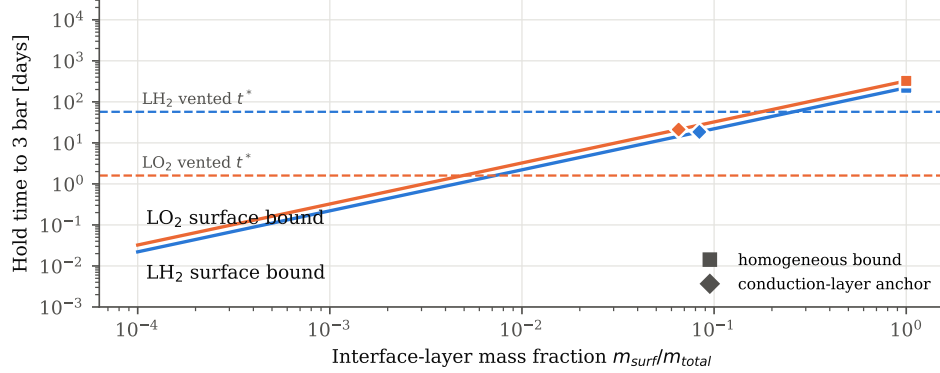


Figure 4: Hold time to 3 bar against interface-layer mass fraction (surface bound), with the homogeneous bound (squares), the conduction-layer anchor (diamonds) and the vented t^* (dashed). Indicative properties.

Table 2: Crossover t^* [days] under each closure (LEO, $r = 2 \text{ m}$, vent at 3 bar). Indicative until the CoolProp run.

	Vented	Surface, $m_{\text{surf}}/m = 10^{-3}$	Surface, conduction anchor	Homogeneous
LH ₂	57.2	57.4	76.8 ($m_{\text{surf}}/m = 0.084$)	409
LO ₂	1.6	2.21	61.7 ($m_{\text{surf}}/m = 0.065$)	943

5 Issues identified

1. **m_{surf} needs a physical anchor.** As $m_{\text{surf}} \rightarrow 0$ the surface bound has $t_{\text{hold}} \rightarrow 0$, so a sweep alone cannot rule out any answer. A reference scale comes from constant-flux conduction into a semi-infinite liquid [6]: $\Delta T_s = (2q/k)\sqrt{\alpha t/\pi}$, which gives $m_{\text{eff}} = \frac{1}{2}\rho A\sqrt{\pi\alpha t}$. The sweep will be reported around this anchor.
2. **The bounds may not collapse cleanly.** The surface closure uses the liquid c_p . The homogeneous closure follows the true constant-volume path, which also includes evaporation into the ullage and liquid expansion. At $m_{\text{surf}} = m$ the two stored energies can differ, and `test_bounds_collapse` is written to measure this. If the difference is significant, the surface layer will be reformulated on the constant-volume saturated path. `test_bounds_ordered` will be run first, because the research claim rests on it.
3. **Fill fraction limits the pressure band.** LH₂ expands strongly on warming. A 90%-full tank may become liquid-full before reaching 3 bar. The (ρ, u) flash captures this, and it will bound the admissible fill fraction in the trade.
4. **Constant k_{eff} and placeholder specific masses** (power, radiator, struts) dominate the uncertainty in the absolute t^* ; all are flagged in Table 3.

6 Parameter register (Stage 1, in progress)

Table 3: Parameter register. P = provisional (must be sourced before publication); S = sourced.

Quantity	Value	Source / status
$T_{\text{sat}}, \rho_l, h_{fg}$ (LH ₂ /LO ₂ /LCH ₄ , 1 atm)	20.3/90.2/111.7 K; 70.8/1141/422 kg/m ³ ; 446/213/511 kJ/kg	CoolProp [2] (reference values)
Cooler mass	$0.2 \dot{Q}^{0.7} [(T_h - T_c)/T_c]^{1.45}$ kg	S [1]
Fraction of Carnot η	0.075 (LH ₂); 0.20 (LO ₂ , LCH ₄)	P
MLI areal density	0.02 kg/m ² per layer	S [1]
MLI k_{eff} , thickness	3×10^{-5} W/m·K; 25 mm per 30 layers	P
Outer cover α/ε	0.32/0.86	P (confirm in [5])
Power system; radiator	0.025 kg/W; 5 kg/m ² , $\varepsilon = 0.9$, 300 K	P
Strut conductance	5×10^{-3} W/K	P

7 Plan to the December milestone

1. **Oct, week 1:** Run the CoolProp test suite and rerun `make_figures.py`; confirm ordering, measure collapse, and replace the indicative closed-tank numbers.
2. **October:** Complete the register (power and radiator specific mass, strut $\int k dT$ for G-10 and Ti, MLI $k_{\text{eff}}(T)$), and reconcile the LO₂ crossover with [1].
3. **October–November:** Port the 2-D solver to L2 (axisymmetric geometry, radiative boundary condition via $h_r = \varepsilon \sigma (T^2 + T_\infty^2)(T + T_\infty)$); add fill fraction and vent pressure to the trade.
4. **November:** Validate against published demonstration-tank boil-off (MHTB [4]) to 10%; check energy closure below 0.5%.
5. **December:** Crossover maps under all three closures; Streamlit interface; public release.

Risk: t^* depends strongly on cooler specific mass, so every map will be published with its sensitivity to s_{cool} , η and s_{rad} .

References

- [1] D. Plachta and P. Kittel, “An updated zero boil-off cryogenic propellant storage analysis applied to upper stages or depots in an LEO environment,” NASA/TM, NTRS 20030067928, 2003.
- [2] I. H. Bell, J. Wronski, S. Quoilin, V. Lemort, “Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp,” *Ind. Eng. Chem. Res.* 53(6):2498–2508, 2014.
- [3] Zumbo, Ferrero, Masseni, Pastrone, “Comparison of thermodynamic modelling methods for cryogenic storage systems,” AIAA SciTech Forum, 2025.
- [4] L. J. Hastings et al., Multipurpose Hydrogen Test Bed (MHTB) MLI and boil-off test reports, NASA MSFC (to be finalised in the register).
- [5] D. G. Gilmore (ed.), *Spacecraft Thermal Control Handbook*, Vol. I, 2nd ed., Aerospace Press, 2002.
- [6] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford, 1959.
- [7] R. F. Barron, *Cryogenic Heat Transfer*, Taylor & Francis, 1999.